



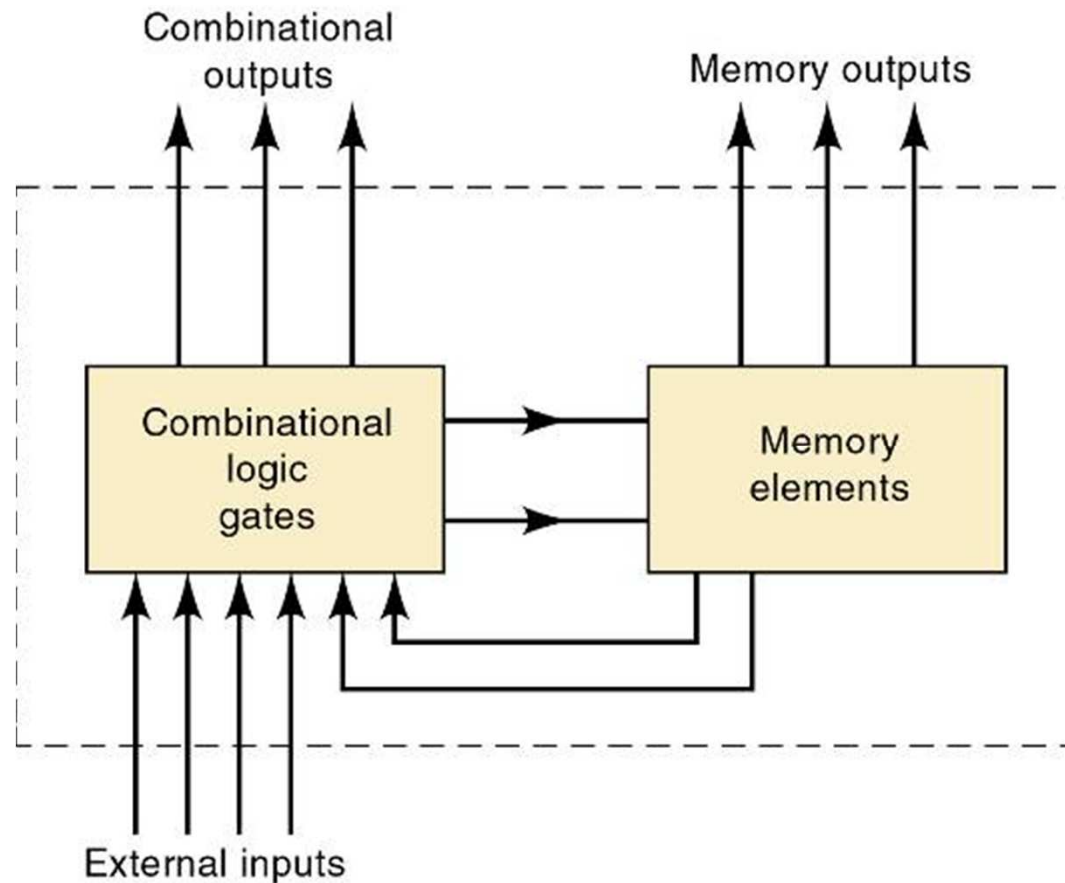
Heaven's Light is Our Guide

Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology, Bangladesh

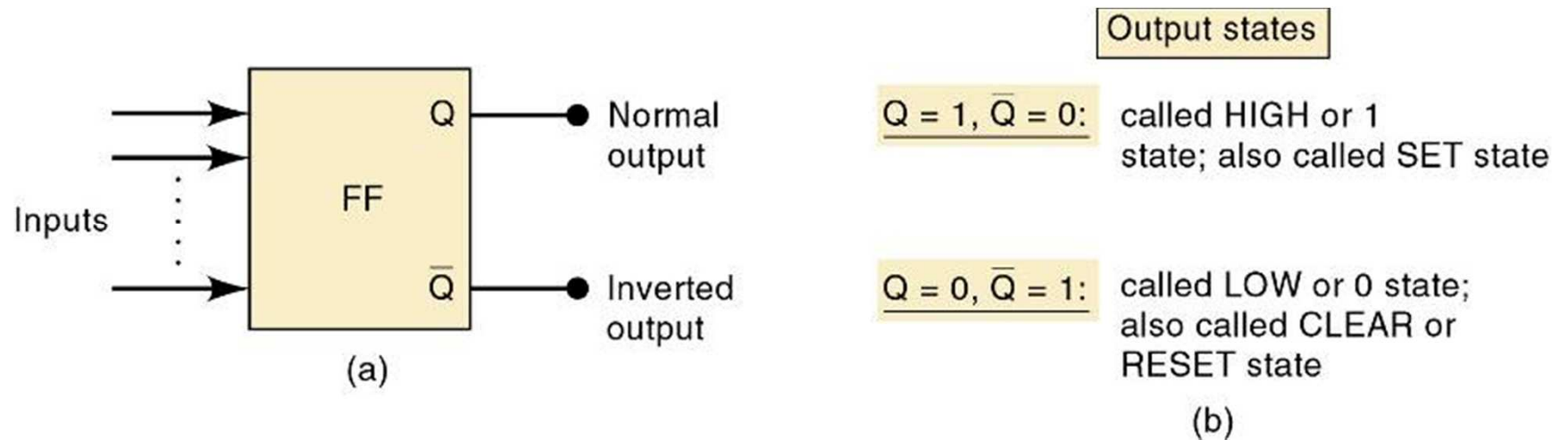
Course Code: CSE 2203
Course Title: Digital Techniques

Presented by,
Prof. Dr. Boshir Ahmed

- Block diagram of a general digital system that combines combinational logic gates with memory devices.



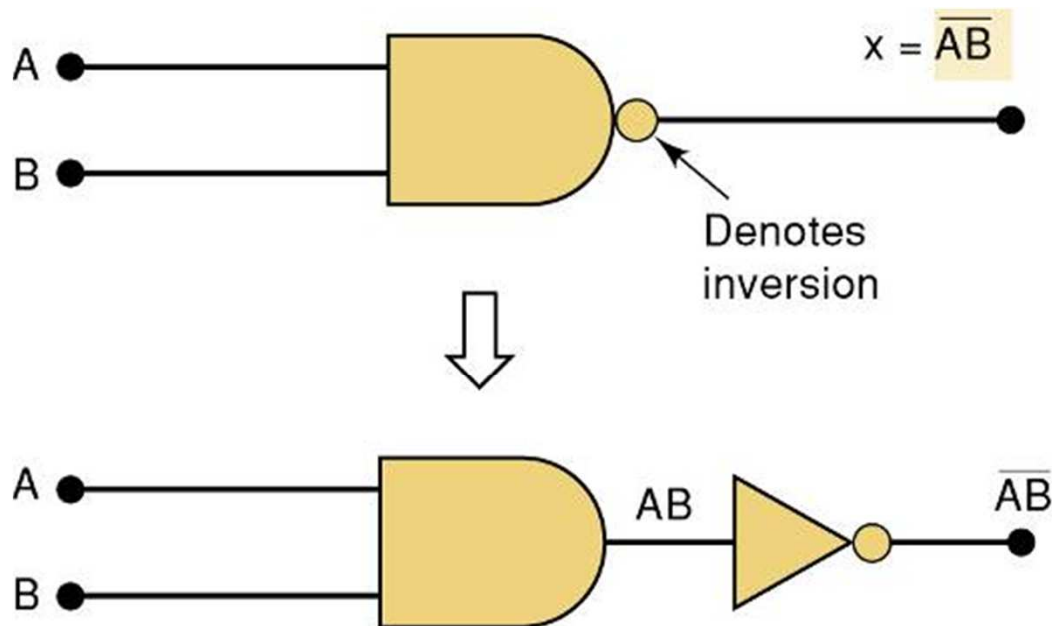
- The most important memory element is the **flip-flop** (FF)—made up of an assembly of logic gates.



The flip-flop is known by other names, including *latch* and *bistable multivibrator*.

NAND Gate Latch

- The NAND gate is an inverted AND gate.
 - An inversion “bubble” is placed at the output of the AND gate, making the Boolean output expression $x = \overline{AB}$

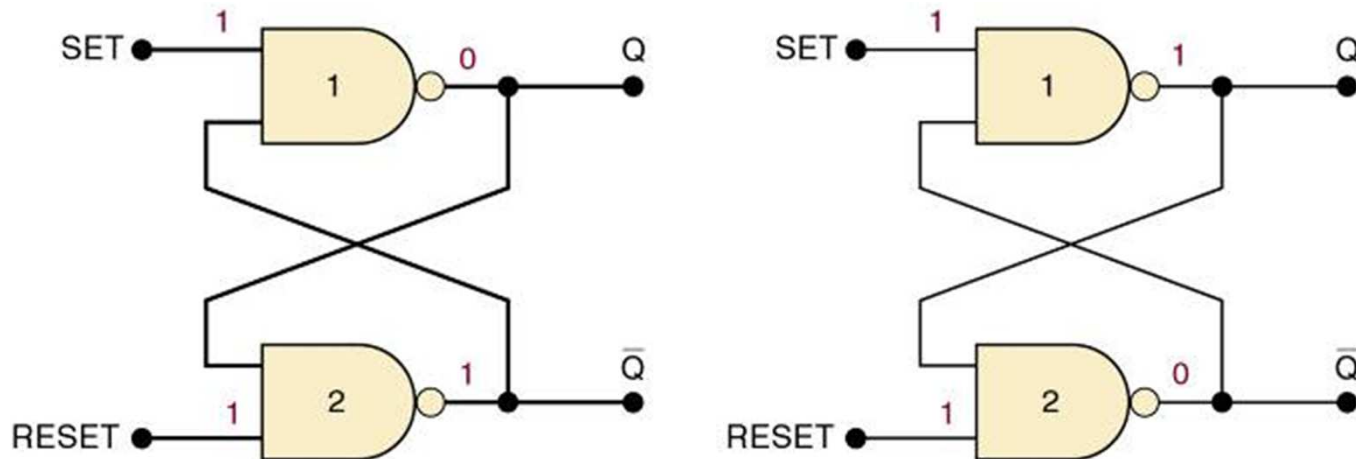


		AND		NAND	
A	B	AB		$\overline{AB}$	
0	0	0		1	
0	1	0		1	
1	0	0		1	
1	1	1		0	

NAND Gate Latch

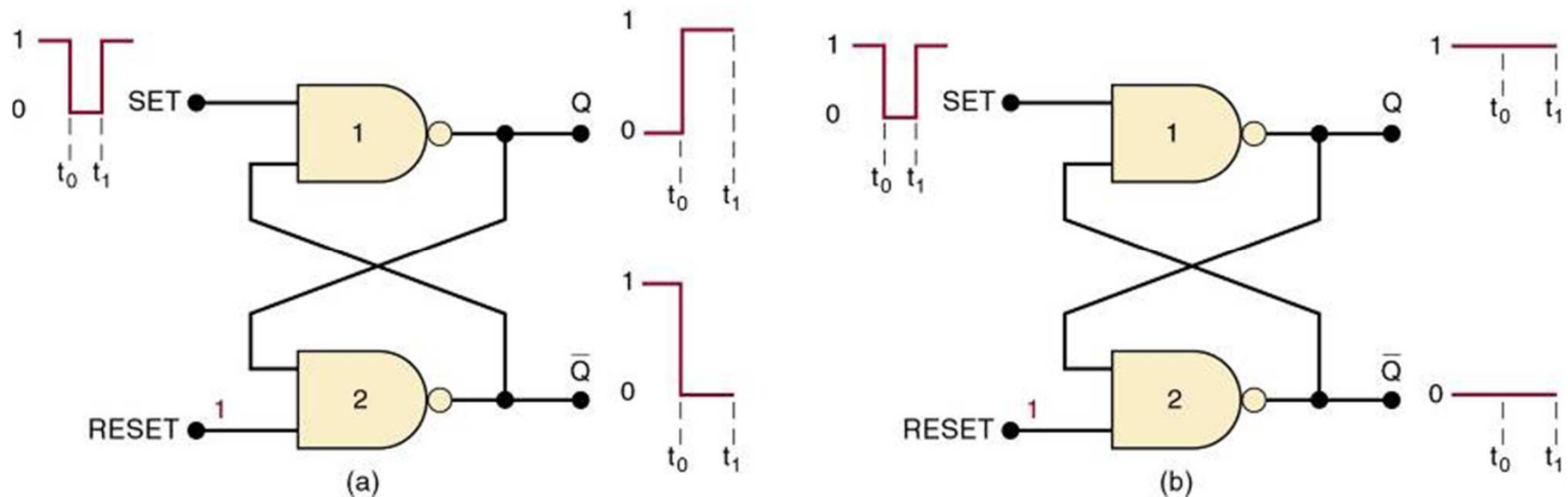
Case - 1

- The **NAND** gate latch or simply latch is a basic FF.
 - Inputs are *SET* and *CLEAR (RESET)*.
- SET = RESET = 1



NAND Gate Latch – Setting the Latch (FF)

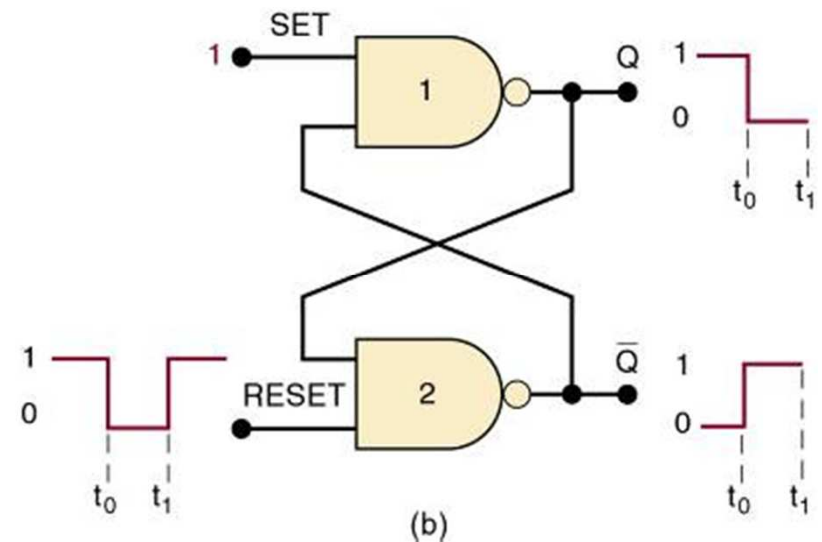
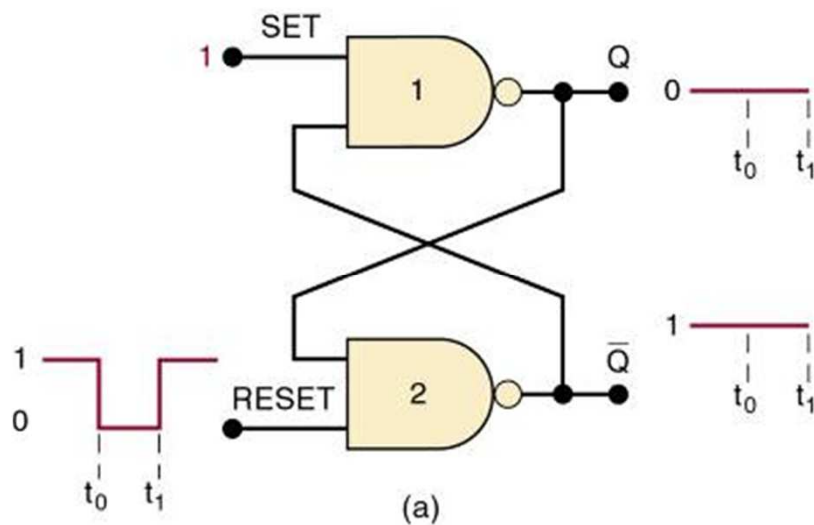
- SET = 0, RESET = 1 state (Case 2)
 - (a) Q = 0 prior to SET pulse.
 - (b) Q = 1 prior to SET pulse.



In both cases, Q ends up HIGH.

NAND Gate Latch – Resetting the Latch (FF)

- SET = 1, RESET = 0 state (**Case 3**)
 - (a) Q = 0 prior to the RESET pulse.
 - (b) Q = 1 prior to the RESET pulse.



In each case, Q ends up LOW.

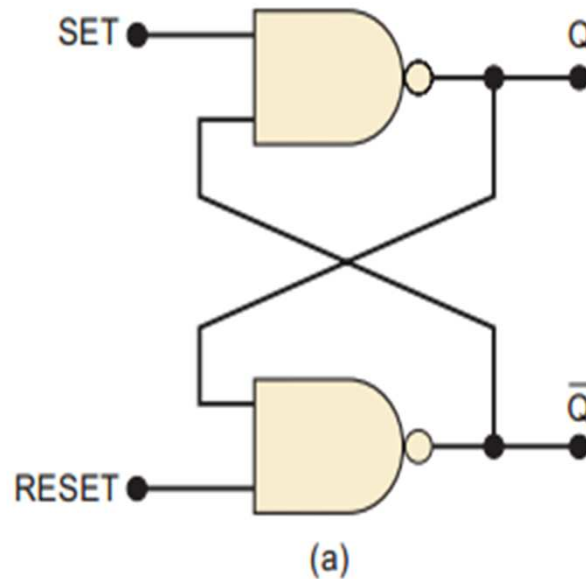
NAND Gate Latch – Resetting the Latch (FF)

- SET = 0, RESET = 0 state (Case 4)
 - Tries to set and clear the latch at the same time, and produces
 - Output is unpredictable, and this input condition should not be used

NAND Gate Latch - Summary

- Summary of the **NAND** latch:
 - **SET = 1, RESET = 1**—Normal resting state, outputs remain in state they were in prior to input.
 - **SET = 0, RESET = 1**—Output will go to $Q = 1$ and remains there, even after SET returns HIGH.
 - Called *setting* the latch.
 - **SET = 1, RESET = 0**—Will produce $Q = 0$ LOW and remains there, even after RESET returns HIGH.
 - Called *clearing* or *resetting* the latch.
 - **SET = 0, RESET = 0**—Tries to set and clear the latch at the same time, and produces $Q = \overline{Q} = 1$.
 - Output is unpredictable, and this input condition should not be used.

NAND Gate Latch Summary



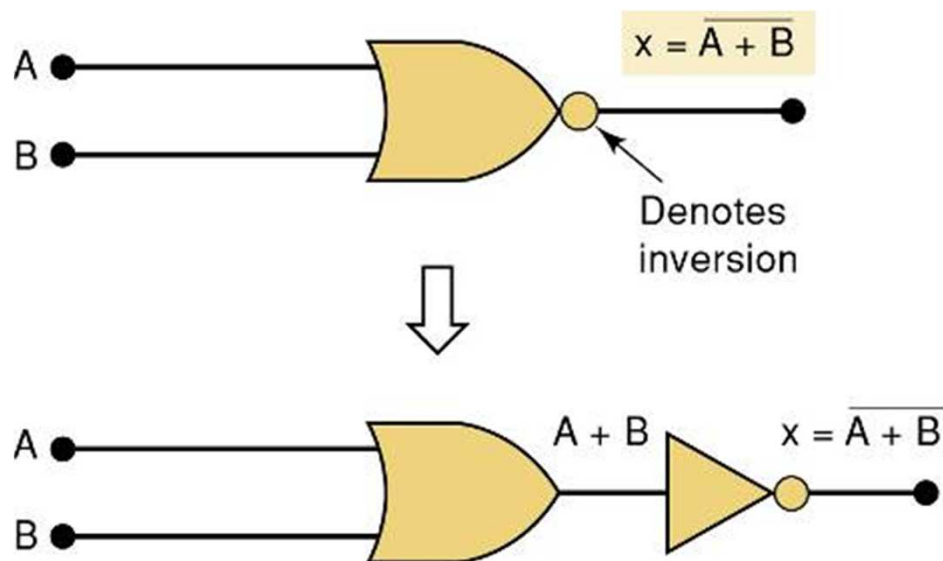
Set	Reset	Output
1	1	No change
0	1	$Q = 1$
1	0	$Q = 0$
0	0	Invalid*

*Produces $Q = \bar{Q} = 1$.

(b)

NOR Gate Latch

- The **NOR** gate is an inverted **OR** gate.
 - An inversion “bubble” is placed at the output of the OR gate, making the Boolean output expression $x = \overline{A + B}$



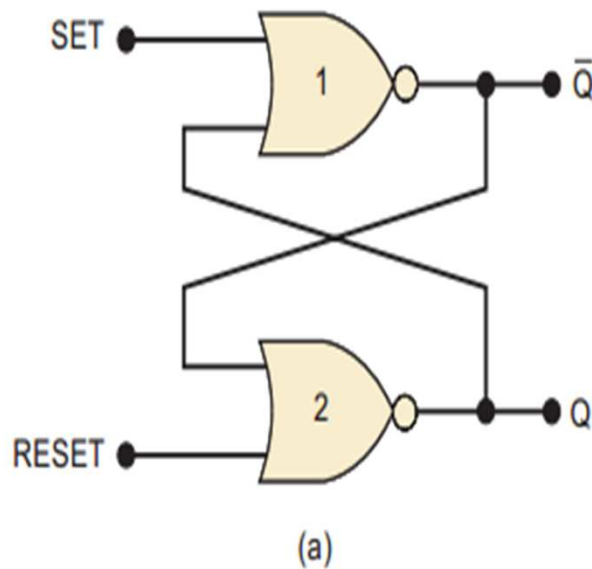
		OR		NOR	
A	B	$A + B$		$\overline{A + B}$	
0	0	0		1	
0	1	1		0	
1	0	1		0	
1	1	1		0	

NOR Gate Latch - Summary

- Summary of the **NAND** latch:
 - **SET = 0, RESET = 0**—Normal resting state, outputs remain in state they were in prior to input.
 - **SET = 1, RESET = 0**—Output will go to $Q = 1$ and remains there, even after SET returns HIGH.
 - Called *setting* the latch.
 - **SET = 0, RESET = 1**—Will produce $Q = 0$ LOW and remains there, even after RESET returns HIGH.
 - Called *clearing* or *resetting* the latch.
 - **SET = 1, RESET = 1**—Tries to set and clear the latch at the same time, and produces $Q = \overline{Q} = 1$.
 - Output is unpredictable, and this input condition should not be used.

NOR Gate Latch Summary

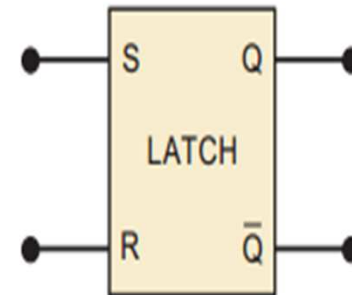
In the next topics, assuming all flip-flops are made of by NOR Latch



Set	Reset	Output
0	0	No change
1	0	$Q = 1$
0	1	$Q = 0$
1	1	Invalid*

*Produces $Q = \bar{Q} = 0$.

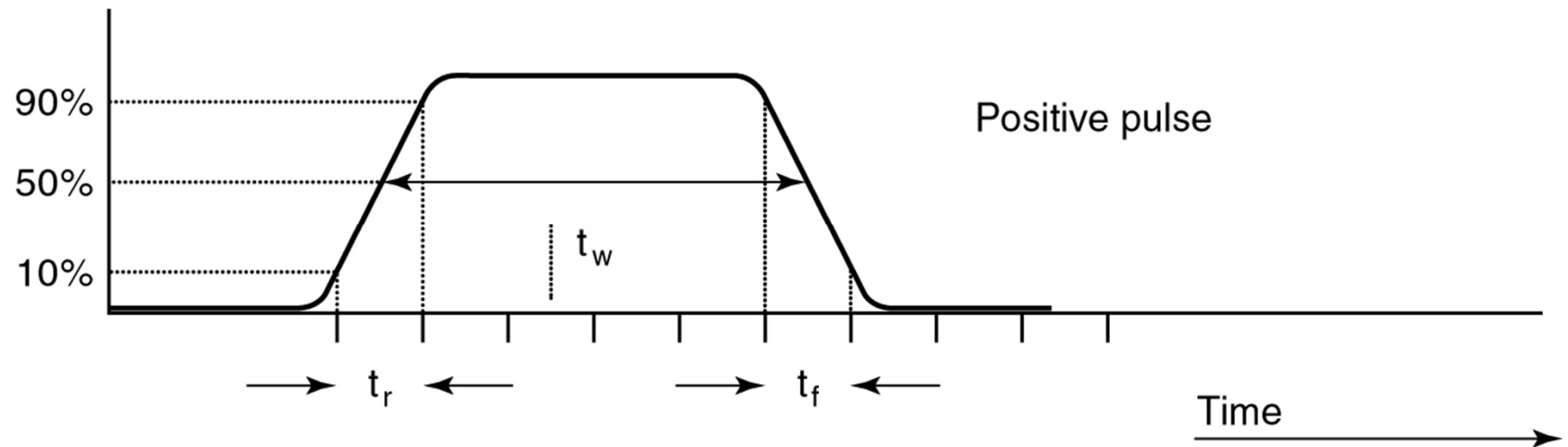
(b)



(c)

Digital Pulses

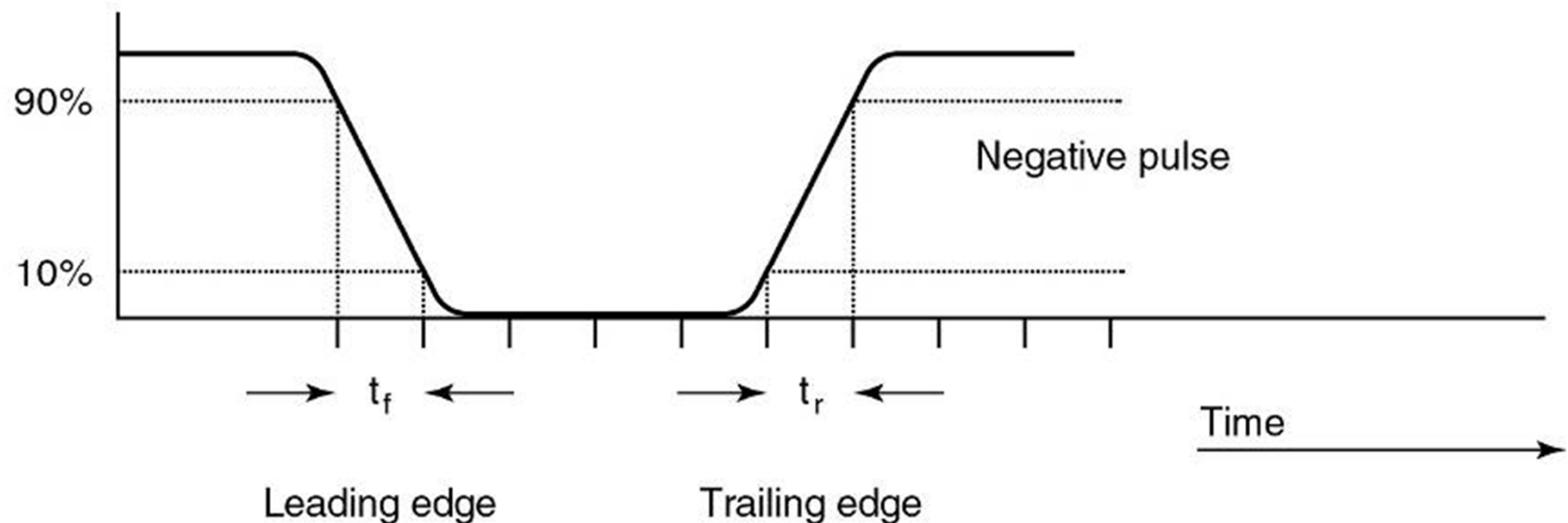
Signals that switch between active and inactive states are called pulse waveforms.



A positive pulse has an active-HIGH level.

Digital Pulses

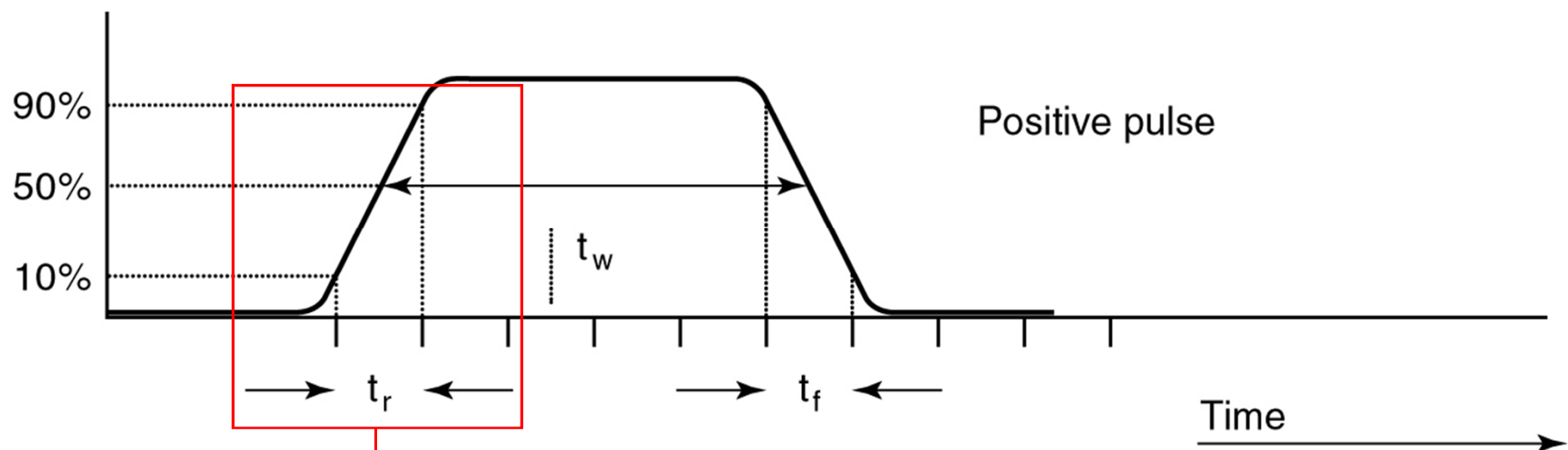
Signals that switch between active and inactive states are called pulse waveforms.



A negative pulse has an active-LOW level.

Digital Pulses

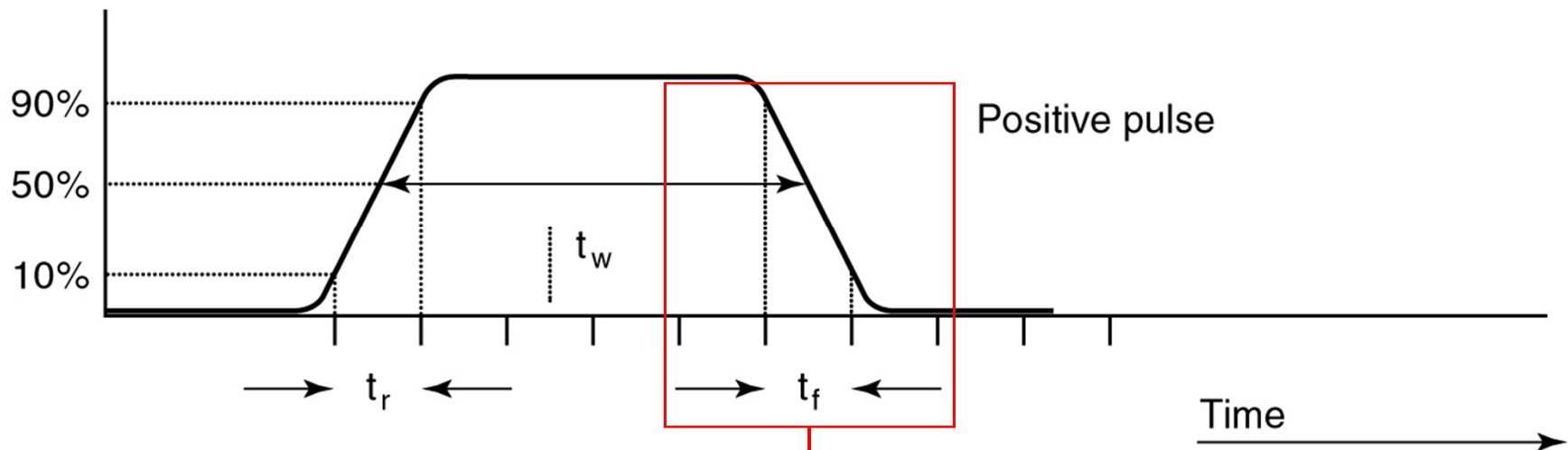
- In actual circuits it takes time for a pulse waveform to change from one level to the other.
 - Transition from LOW to HIGH on a positive pulse is called *rise time* (t_r).



Measured between the 10% and 90% points on the *leading edge* of the voltage waveform.

Digital Pulses

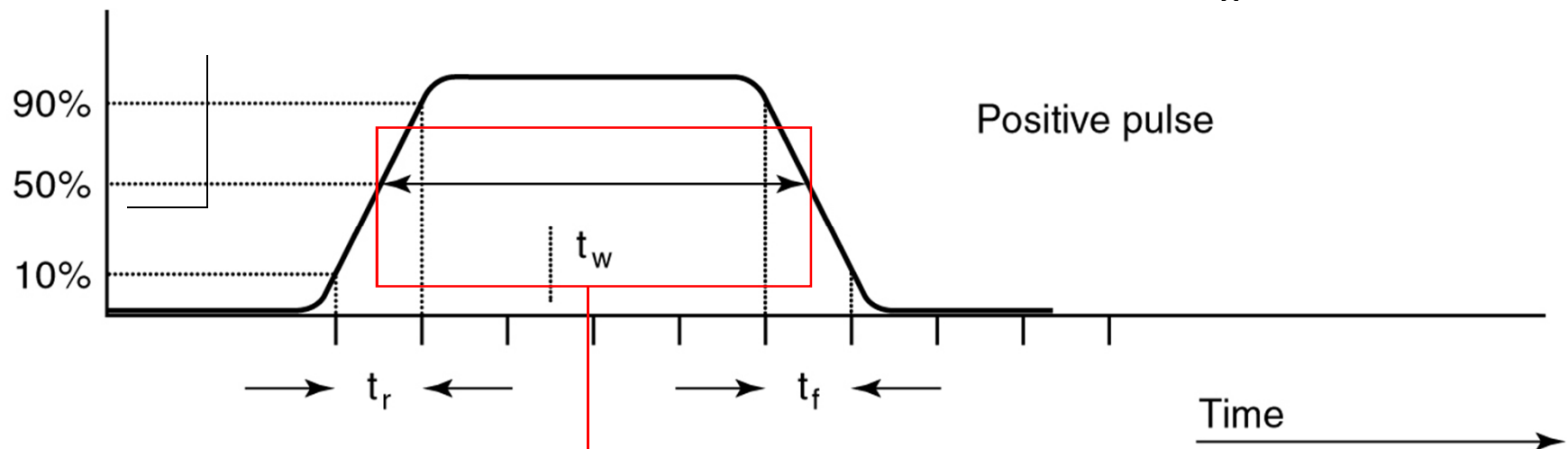
- In actual circuits it takes time for a pulse waveform to change from one level to the other.
 - Transition from HIGH to LOW on a positive pulse is called *fall time* (t_f).



Measured between the 90% and 10% points on the *trailing edge* of the voltage waveform.

Digital Pulses

- In actual circuits it takes time for a pulse waveform to change from one level to the other.
 - A pulse also has a *duration*—width—(t_w).



The time between the points when the leading and trailing edges are at 50% of the HIGH level voltage.

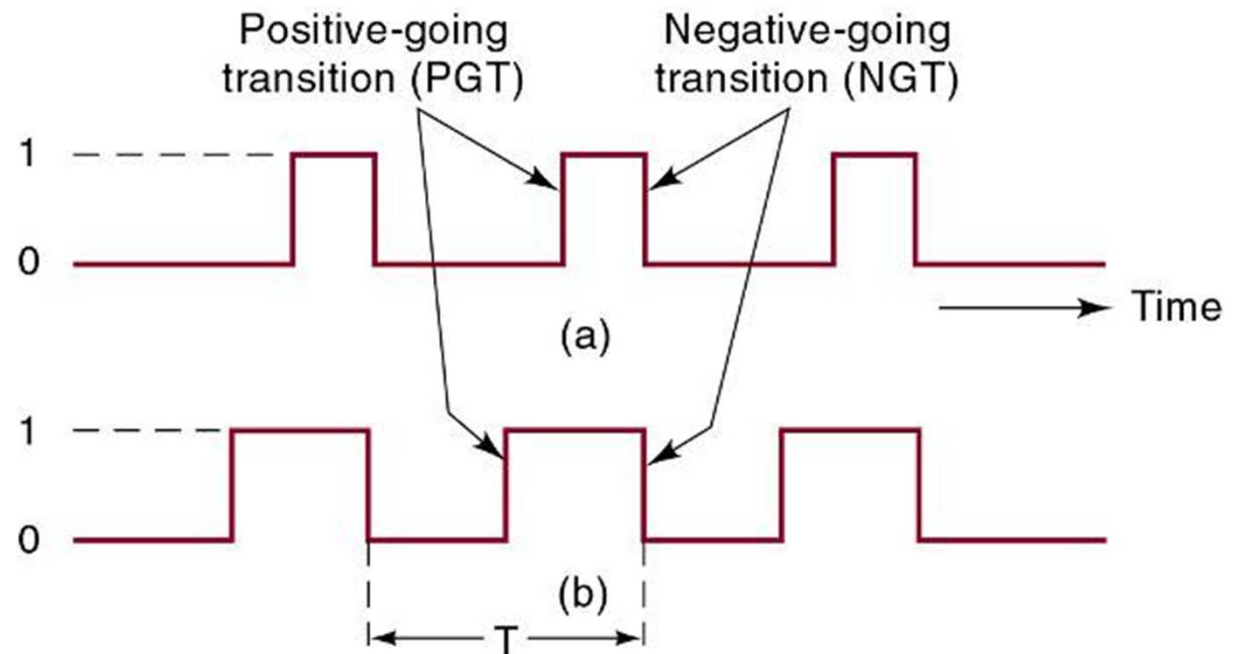
Clock Signals and Clocked Flip-Flops

- Digital systems can operate either *asynchronously* or *synchronously*.
 - **Asynchronous system**—outputs can change state at any time the input(s) change.
 - **Synchronous system**—output can change state only at a specific time in the clock cycle.

Clock Signals and Clocked Flip-Flops

- The clock signal is a rectangular pulse train or square wave.
 - Positive going transition (PGT)—clock pulse goes from 0 to 1.
 - Negative going transition (NGT)—clock pulse goes from 1 to 0.

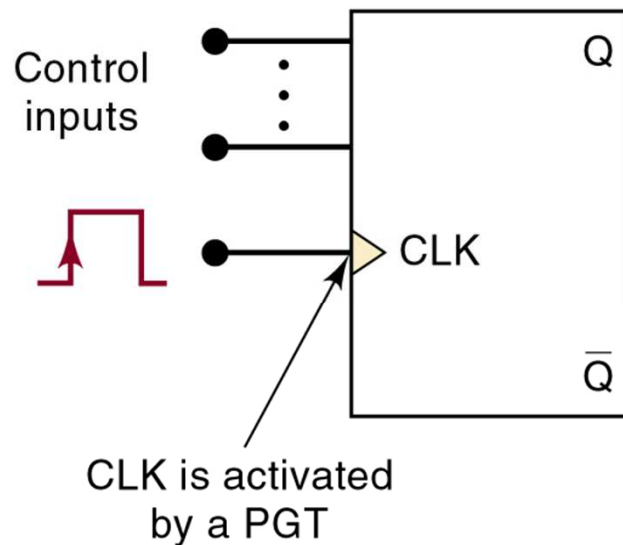
Transitions are also called *edges*.



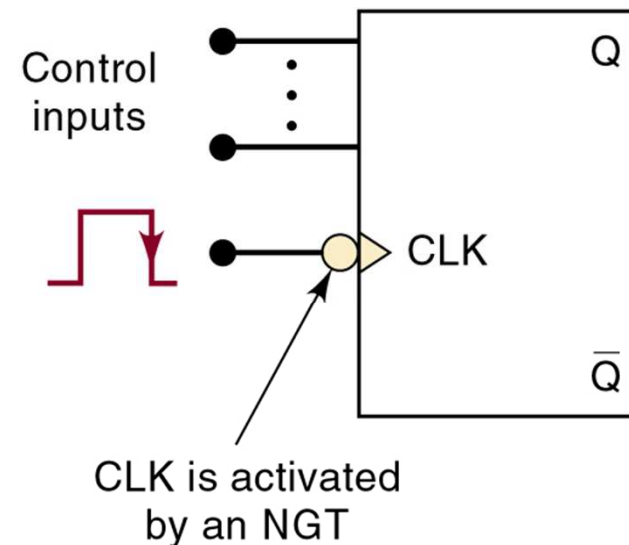
Clock Signals and Clocked Flip-Flops

- Clocked FFs change state on one or the other clock transitions.
 - Clock inputs are labeled CLK, CK, or CP.

A small triangle at the CLK input indicates that the input is activated with a PGT.

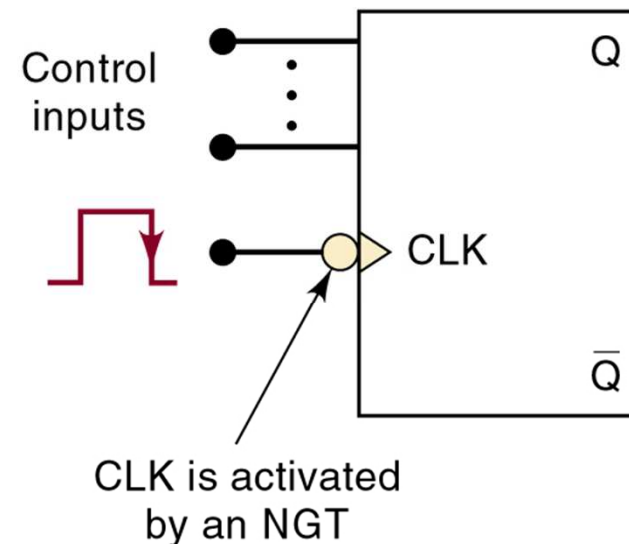
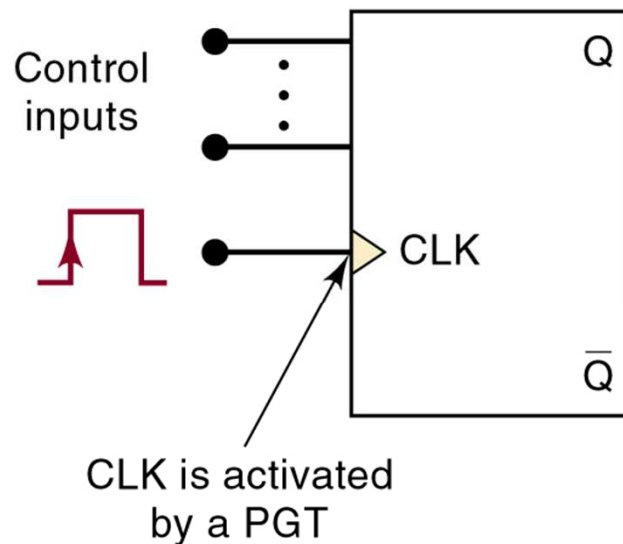


A bubble and a triangle indicates that the CLK input is activated with a NGT.



Clock Signals and Clocked Flip-Flops

- Control inputs have an effect on the output only at the active clock transition (NGT or PGT)—also called synchronous control inputs.
 - The control inputs get the outputs ready to change, but the change is not triggered until the CLK edge.

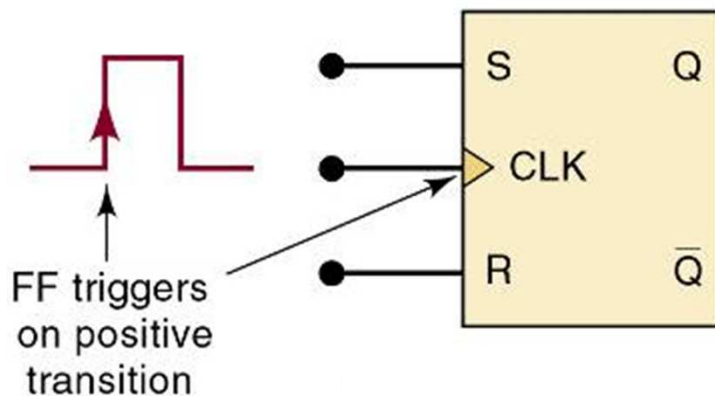


Clocked S-R Flip-Flop

- The S and R inputs are synchronous *control* inputs, which control the state the FF will go to when the clock pulse occurs.
 - The CLK input is the **trigger** input that causes the FF to change states according to the S and R inputs.
- SET-RESET (or SET-CLEAR) FF will change states at positive- or negative-going clock edges.

Clocked S-R Flip-Flop

A clocked S-R flip-flop triggered by the positive-going edge of the clock signal.



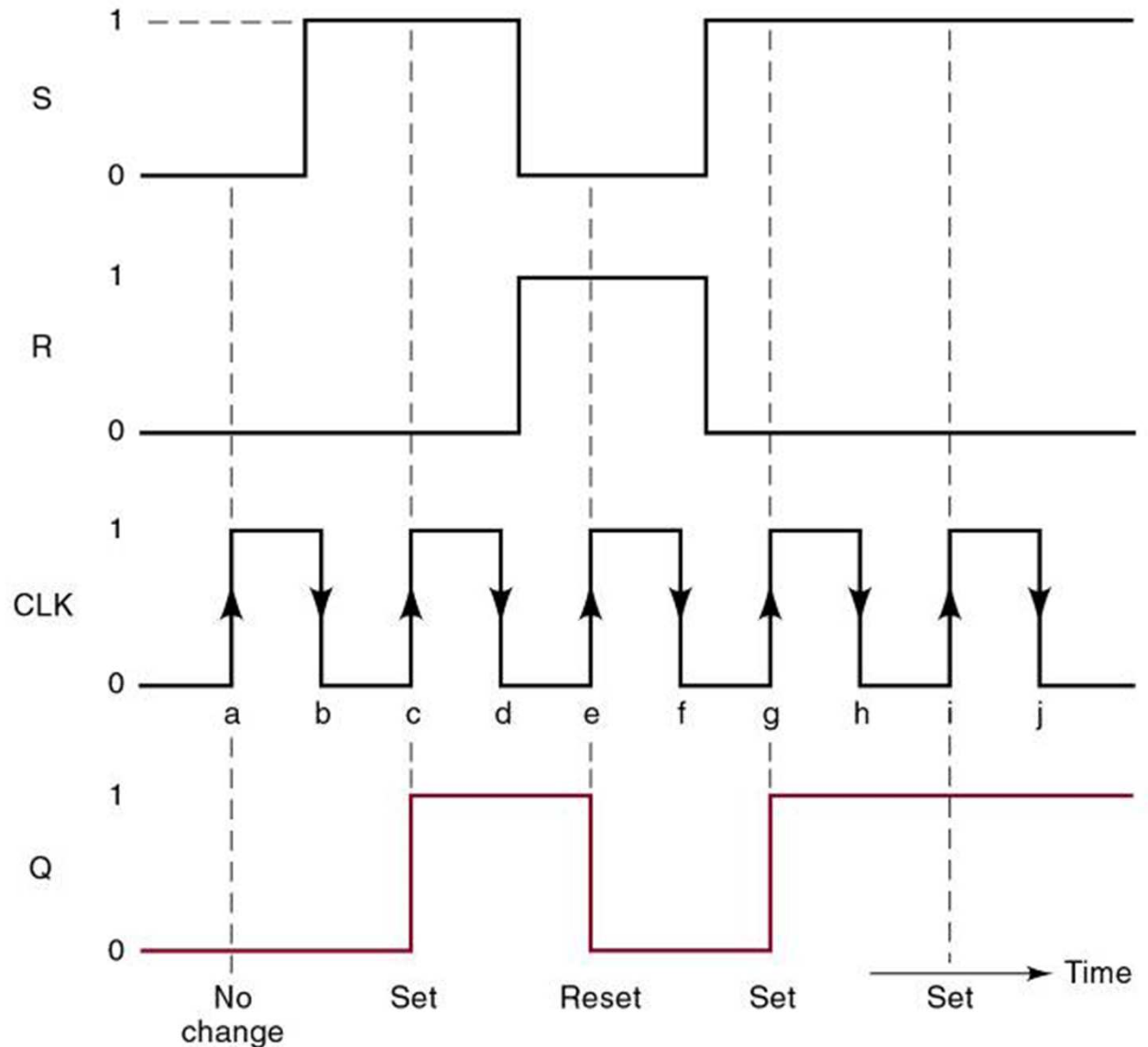
Inputs			Output
S	R	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	Ambiguous

Q_0 is output level prior to ↑ of CLK.
↓ of CLK produces no change in Q.

The S and R inputs control the state of the FF in the same manner as described earlier for the NOR gate latch, but the FF does *not* respond to these inputs *until* the occurrence of the PGT of the clock signal.

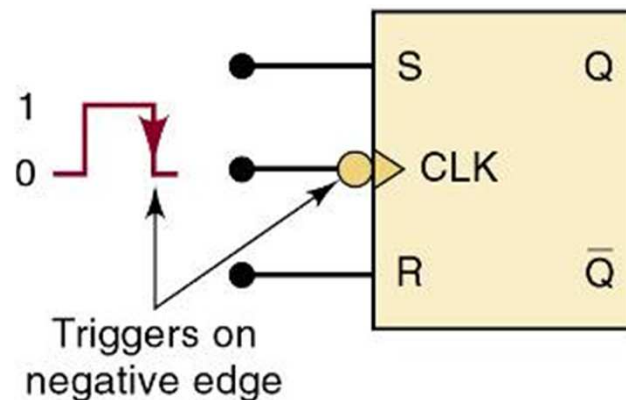
Clocked S-R Flip-Flop

Waveforms of the operation of a clocked S-R flip-flop triggered by the positive-going edge of a clock pulse.



Clocked S-R Flip-Flop

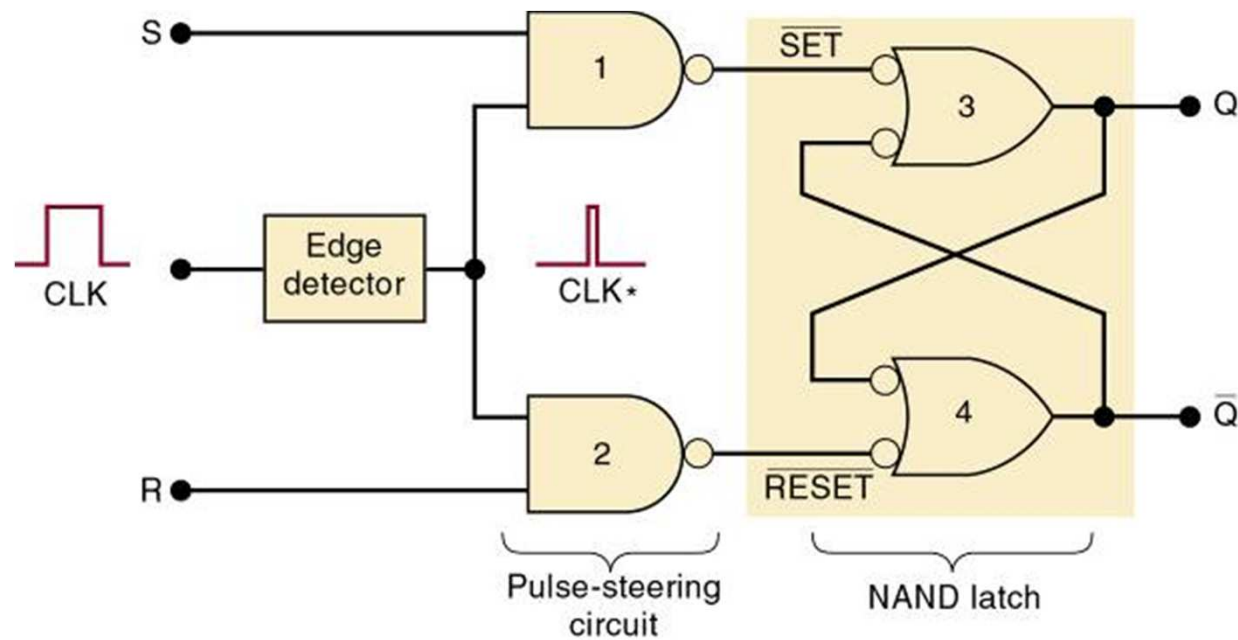
A clocked S-R flip-flop triggered by the negative-going edge of the clock signal.



Inputs			Output
S	R	CLK	Q
0	0	↓	Q_0 (no change)
1	0	↓	1
0	1	↓	0
1	1	↓	Ambiguous

Both positive-edge and negative-edge triggering FFs are used in digital systems.

Clocked S-R Flip-Flop – Internal Circuitry



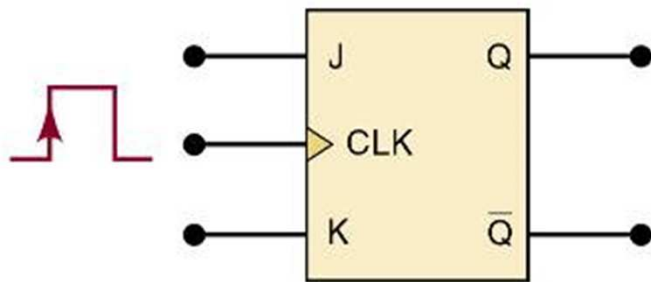
- An edge-triggered S-R flip-flop circuit features:
 - A basic **NAND** gate latch formed by **NAND-3** and **NAND-4**.
 - A **pulse-steering circuit** formed by **NAND-1** and **NAND-2**.
 - An **edge-detector circuit**.

Clocked J-K Flip-Flop

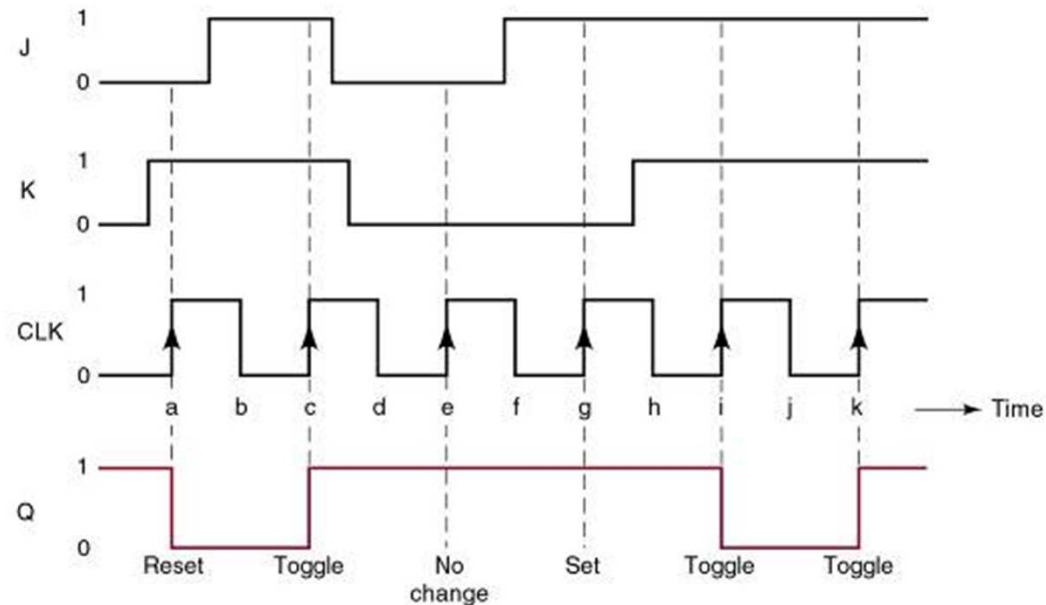
- Operates like the S-R FF.
 - J is SET, K is CLEAR.
- When J and K are both HIGH, output is toggled to the opposite state.
- In toggle state, output will go just opposite state.
 - May be positive going or negative going clock trigger.
- Much more versatile than the S-R flip-flop, as it has no ambiguous states.
 - Has the ability to do everything the S-R FF does, plus operates in toggle mode.

Clocked J-K Flip-Flop

Clocked J-K flip-flop that responds only to the positive edge of the clock.

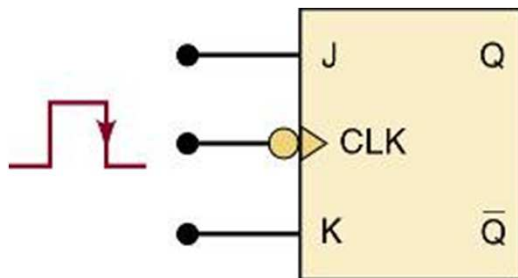


J	K	CLK	Q
0	0	$\uparrow$	Q_0 (no change)
1	0	$\uparrow$	1
0	1	$\uparrow$	0
1	1	$\uparrow$	$\bar{Q}_0$ (toggles)



Clocked J-K Flip-Flop

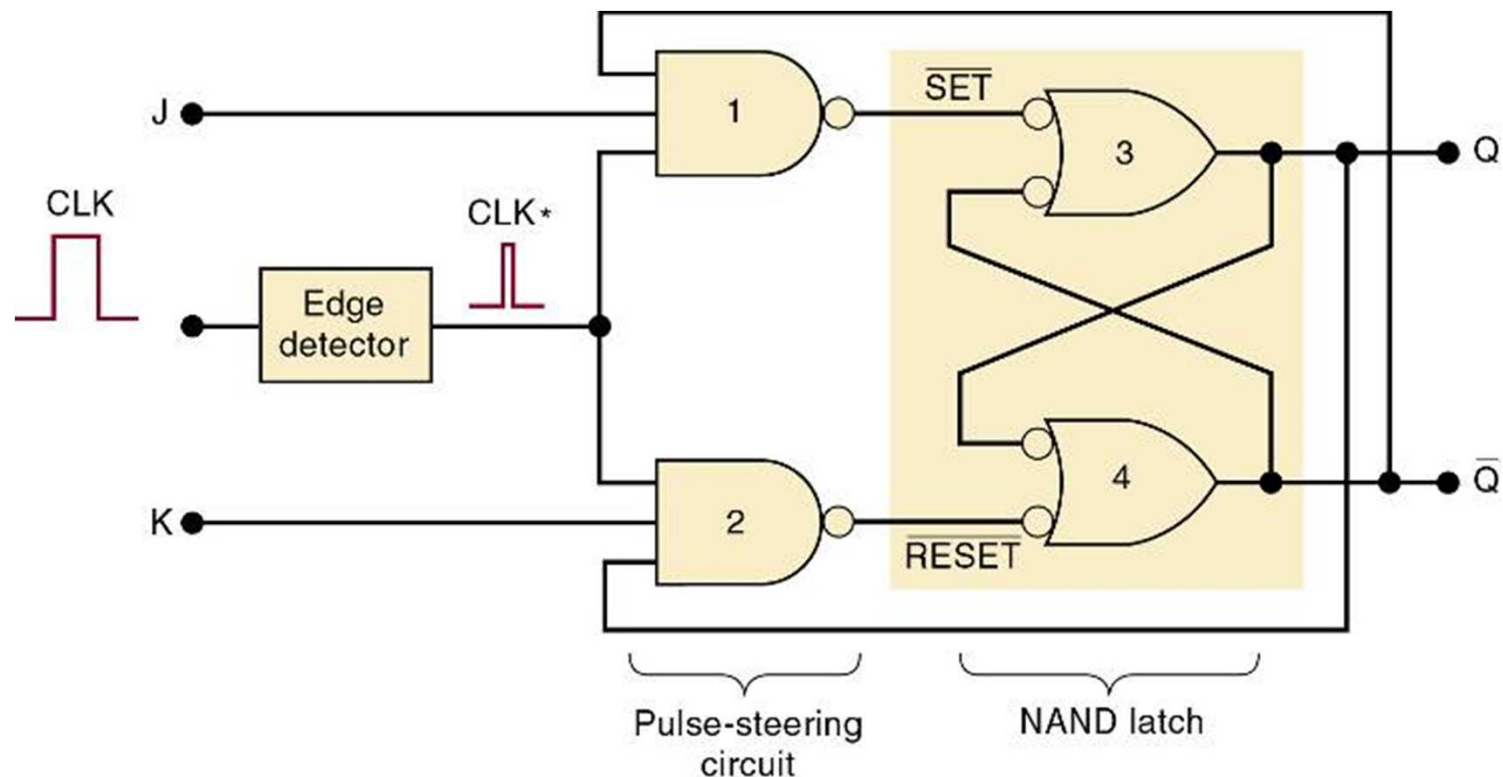
Clocked J-K flip-flop that responds only to the negative edge of the clock.



J	K	CLK	Q
0	0	↓	Q_0 (no change)
1	0	↓	1
0	1	↓	0
1	1	↓	$\bar{Q}_0$ (toggles)

Clocked J-K Flip-Flop – Internal Circuitry

- The internal circuitry of an edge-triggered J-K flip-flop contains the same three sections as the edge-triggered S-R flip-flop.

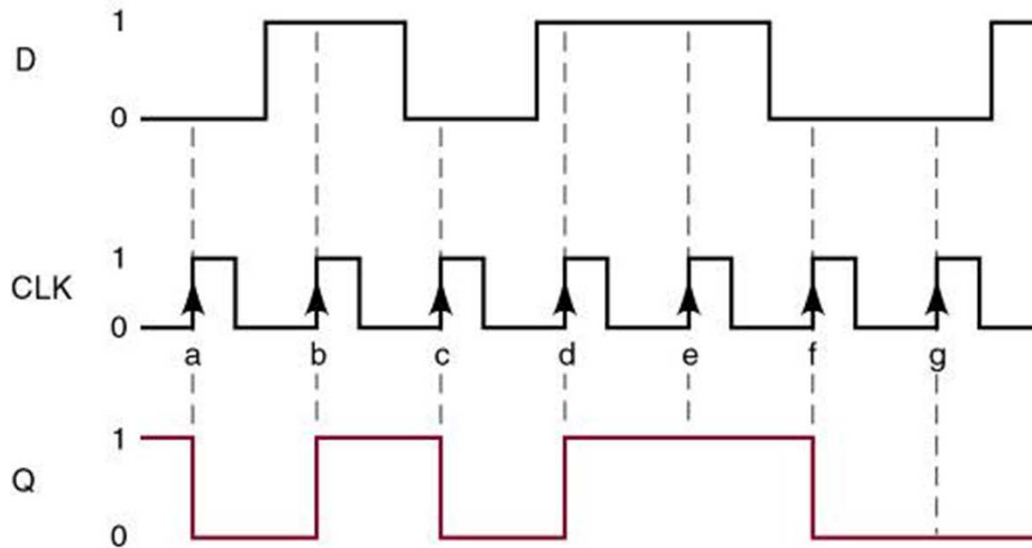
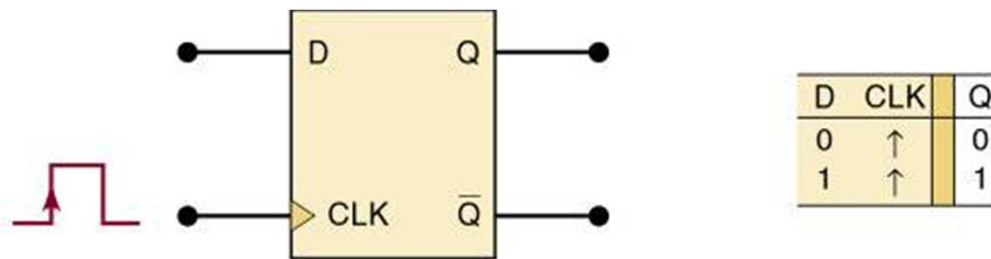


Clocked D Flip-Flop

- One data input—output changes to the value of the input at either the positive- or negative-going clock trigger.
- May be implemented with a J-K FF by tying the J input to the K input through an inverter.
- By setting $J = D$, $K = \neg D$ in JK flip-flop D flip-flop can be implemented
- Useful for parallel data transfer.

Clocked D Flip-Flop

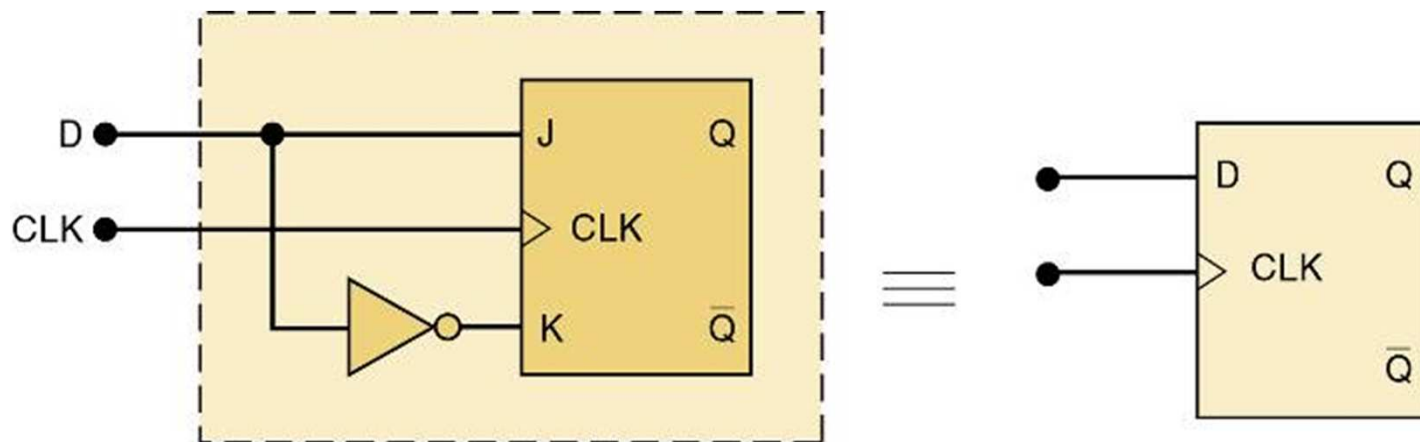
D flip-flop that triggers only on positive-going transitions.



Clocked D Flip-Flop - Implementation

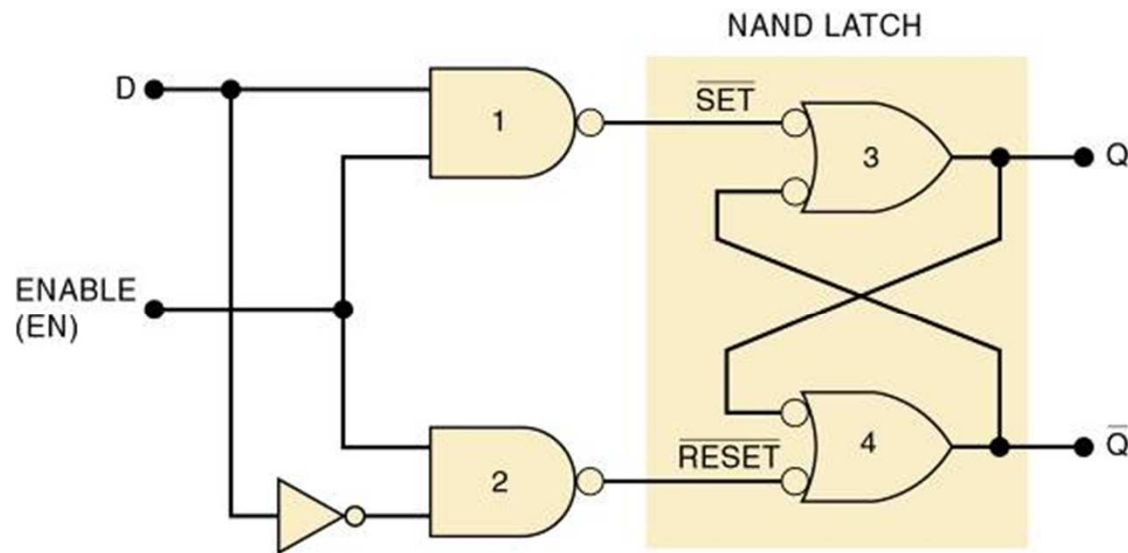
- An edge-triggered D flip-flop is implemented by adding a single INVERTER to the edge-triggered J-K flip-flop.
 - The same can be done to convert a S-R flip-flop to a D flip-flop.

Edge-triggered D flip-flop implementation from a J-K flip-flop.



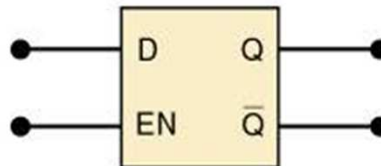
D Latch (Transparent Latch)

D latch structure, function table, logic symbol.



Inputs		Output
EN	D	Q
0	X	Q_0 (no change)
1	0	0
1	1	1

"X" indicates "don't care."
 Q_0 is state Q just prior to EN going LOW.



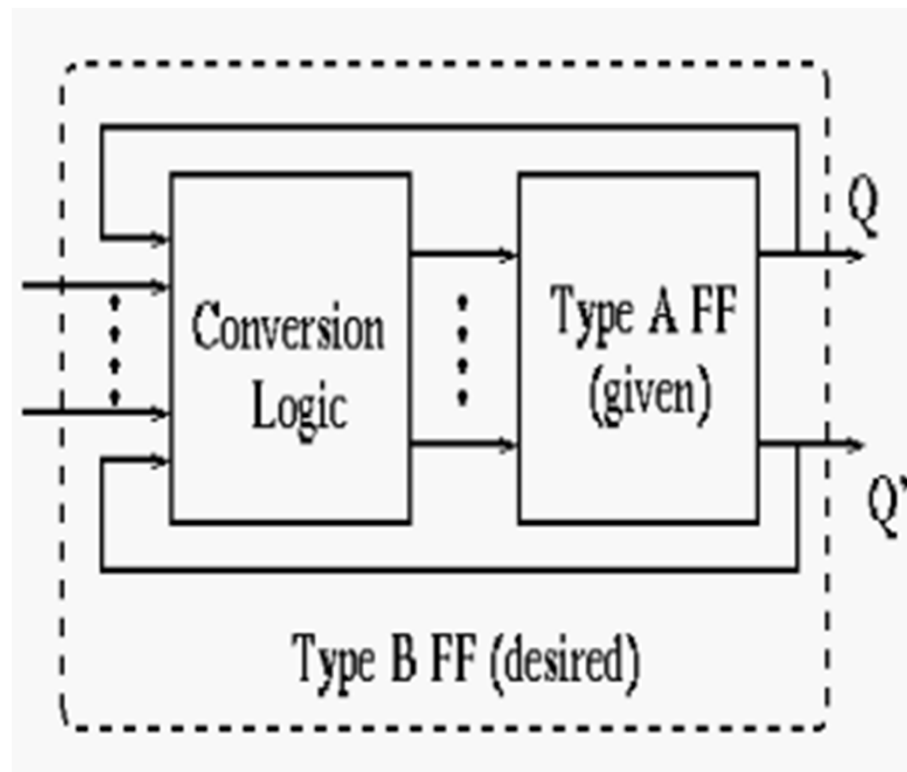
Clocked T Flip-Flop

T flip-flop can be obtained by setting **J = T** and **K = T** in JK flip-flop.

T	CLK	Output (Q)
0	↑	Q_0 (no change)
1	↑	toggle

Flipflop Conversions

- The purpose is to convert a given type A FF to a desired type B FF using some conversion logic.



Excitation Table

The key here is to use the excitation table, which shows the necessary triggering signal (SR, JK, D and T) for a desired flip flop state transition $Q_t - Q_{t+1}$:

Q_t	Q_{t+1}	S	R	J	K	D	T
0	0	0	x	0	x	0	0
0	1	1	0	1	x	1	1
1	0	0	1	x	1	0	1
1	1	x	0	x	0	1	0

Excitation Table Generation Process

- SR Flip Flop

S	R	Present State Q_t	Next State Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Excitation Table Generation Process

- SR Flip Flop

S	R	Present State Q_t	Next State Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Excitation Table Generation Process

- SR Flip Flop

Present State Q_t	Next State Q_{t+1}	S	R
0	0	0	x

Excitation Table Generation Process

- SR Flip Flop

S	R	Present State Q_t	Next State Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Excitation Table Generation Process

- SR Flip Flop

Present State Q_t	Next State Q_{t+1}	S	R
0	0	0	x
0	1	1	0

Excitation Table Generation Process

- SR Flip Flop

S	R	Present State Q_t	Next State Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Excitation Table Generation Process

- SR Flip Flop

Present State Q_t	Next State Q_{t+1}	S	R
0	0	0	×
0	1	1	0
1	0	0	1

Excitation Table Generation Process

- SR Flip Flop

S	R	Present State Q_t	Next State Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Excitation Table Generation Process

- SR Flip Flop

Present State Q_t	Next State Q_{t+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

Excitation Table Generation Process

- JK Flip Flop

Please, Prepare the Excitation Table of JK Flip Flop as SR Flip Flop in Home.

Excitation Table Generation Process

- JK Flip Flop (After preparing, you will see as the following table)

Present State Q_t	Next State Q_{t+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Excitation Table Generation Process

- D Flip Flop

D	Present State Q_t	Next State Q_{t+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table Generation Process

- D Flip Flop

D	Present State Q_t	Next State Q_{t+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table Generation Process

- D Flip Flop

Present State Q_t	Next State Q_{t+1}	D
0	0	0

Excitation Table Generation Process

- D Flip Flop

D	Present State Q_t	Next State Q_{t+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table Generation Process

- D Flip Flop

Present State Q_t	Next State Q_{t+1}	D
0	0	0
0	1	1

Excitation Table Generation Process

- D Flip Flop

D	Present State Q_t	Next State Q_{t+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table Generation Process

- D Flip Flop

Present State Q_t	Next State Q_{t+1}	D
0	0	0
0	1	1
1	0	0

Excitation Table Generation Process

- D Flip Flop

D	Present State Q_t	Next State Q_{t+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table Generation Process

- D Flip Flop

Present State Q_t	Next State Q_{t+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table Generation Process

- T Flip Flop

Please, Prepare the Excitation Table of T Flip Flop as D Flip Flop in Home.

Excitation Table Generation Process

- T Flip Flop (After preparing, you will see as the following table)

Present State Q_t	Next State Q_{t+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

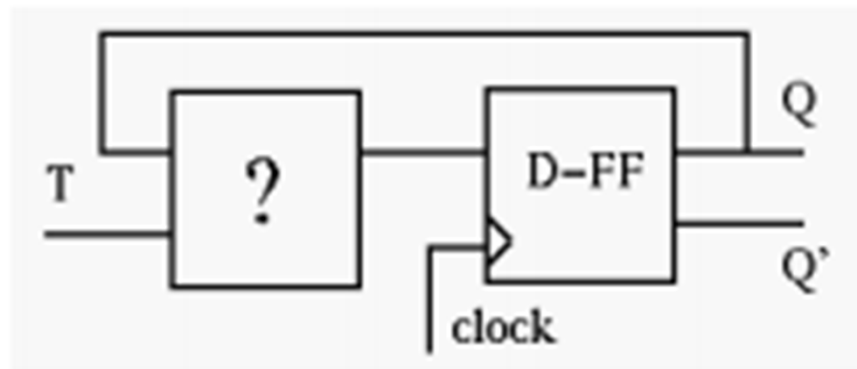
Excitation Table

The key here is to use the excitation table, which shows the necessary triggering signal (SR, JK, D and T) for a desired flip flop state transition $Q_t - Q_{t+1}$:

Q_t	Q_{t+1}	S	R	J	K	D	T
0	0	0	x	0	x	0	0
0	1	1	0	1	x	1	1
1	0	0	1	x	1	0	1
1	1	x	0	x	0	1	0

Convert a D-FF to a T-FF

The output of D flip flop should be as the output of T flip flop. We need to design the circuit to generate the triggering signal D as a function of T and Q: $D = f(T, Q)$



Consider the excitation table of T and D Flip flops. Write Down Excitation Table of T, Q_n and Q_{n+1} , D. For the K-map, consider T and Q_n as Input and D as output.
 $D = TQ_n' + T'Q_n$ (Ex- OR gate).

T	Q_n	Q_{n+1}	D
0	0	0	0
1	0	1	1
1	1	0	0
0	1	1	1

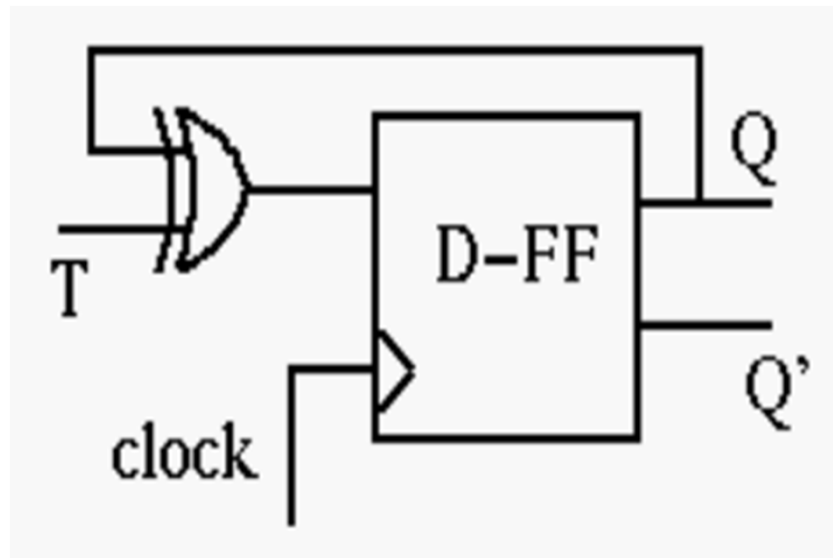
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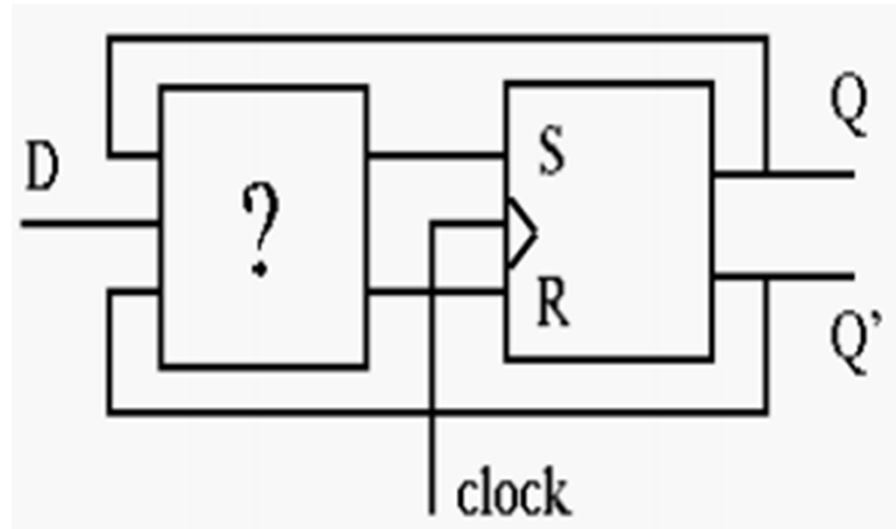
Convert a D-FF to a T-FF

- Treating as a function of and current FF state Q (Q_t), we have:

$$D = T'Q + TQ' = T \oplus Q$$



Convert a RS-FF to a D-FF



We need to design the circuit to generate the triggering signals S and R as functions of D and Q.

Consider the excitation table

D	Q_t	Q_{t+1}	S	R
0	0	0	0	x
1	0	1	1	0
0	1	0	0	1
1	1	1	x	0

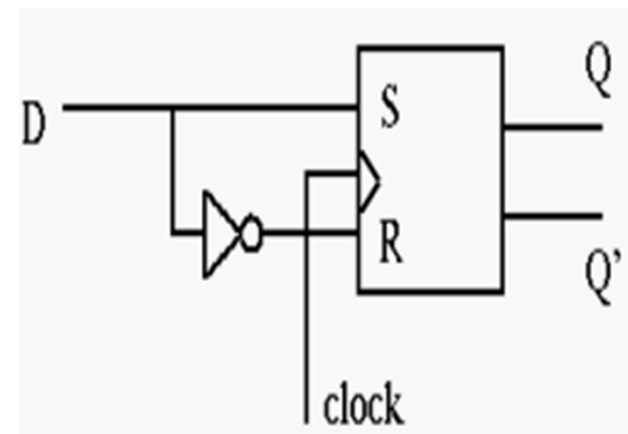
The desired signal S and R can be obtained as functions of D and Q current FF state from the Karnaugh maps:

D \ Q	0	1
0	0	0
1	1	x

$$S = D$$

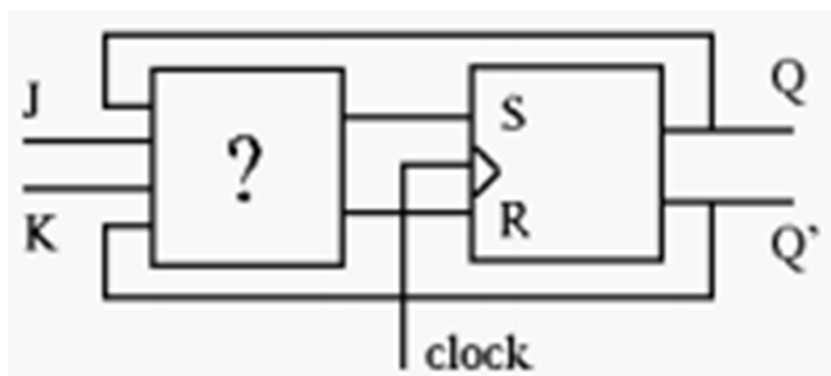
D \ Q	0	1
0	x	1
1	0	0

$$R = D'$$



Convert a RS-FF to a JK-FF

We need to design the circuit to generate the triggering signals S and R as functions of J, K and Q. Consider the excitation table:



J	K	Q_t	Q_{t+1}	S	R
0	x	0	0	0	x
1	x	0	1	1	0
x	1	1	0	0	1
x	0	1	1	x	0

Convert a RS-FF to a JK-FF

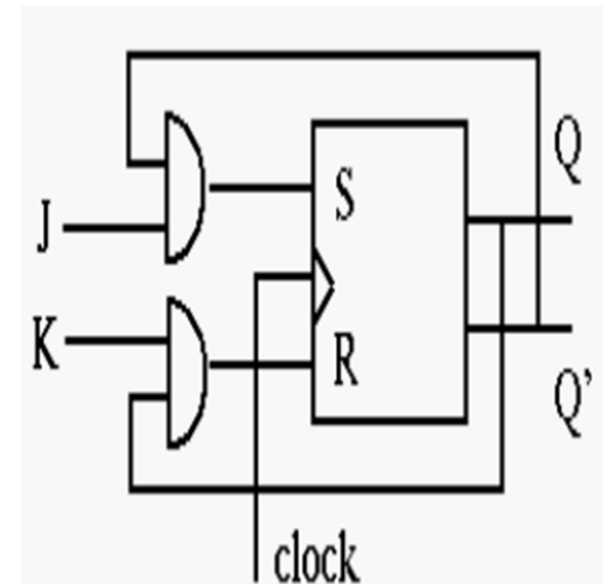
The desired signal S and R as functions of J, K and current FF state Q can be obtained from the Karnaugh maps:

K \ QJ	00	01	11	10
	0	1	X	X
1	0	1	0	0

$$S = Q'J$$

K \ QJ	00	01	11	10
	X	0	0	0
1	X	0	1	1

$$R = QK$$



Flip-Flop Applications

- Examples of applications:
 - Counting; Storing binary data
 - Transferring binary data between locations
- Many FF applications are categorized sequential.
 - Output follows a predetermined sequence of states.

Thank You



Next topic